

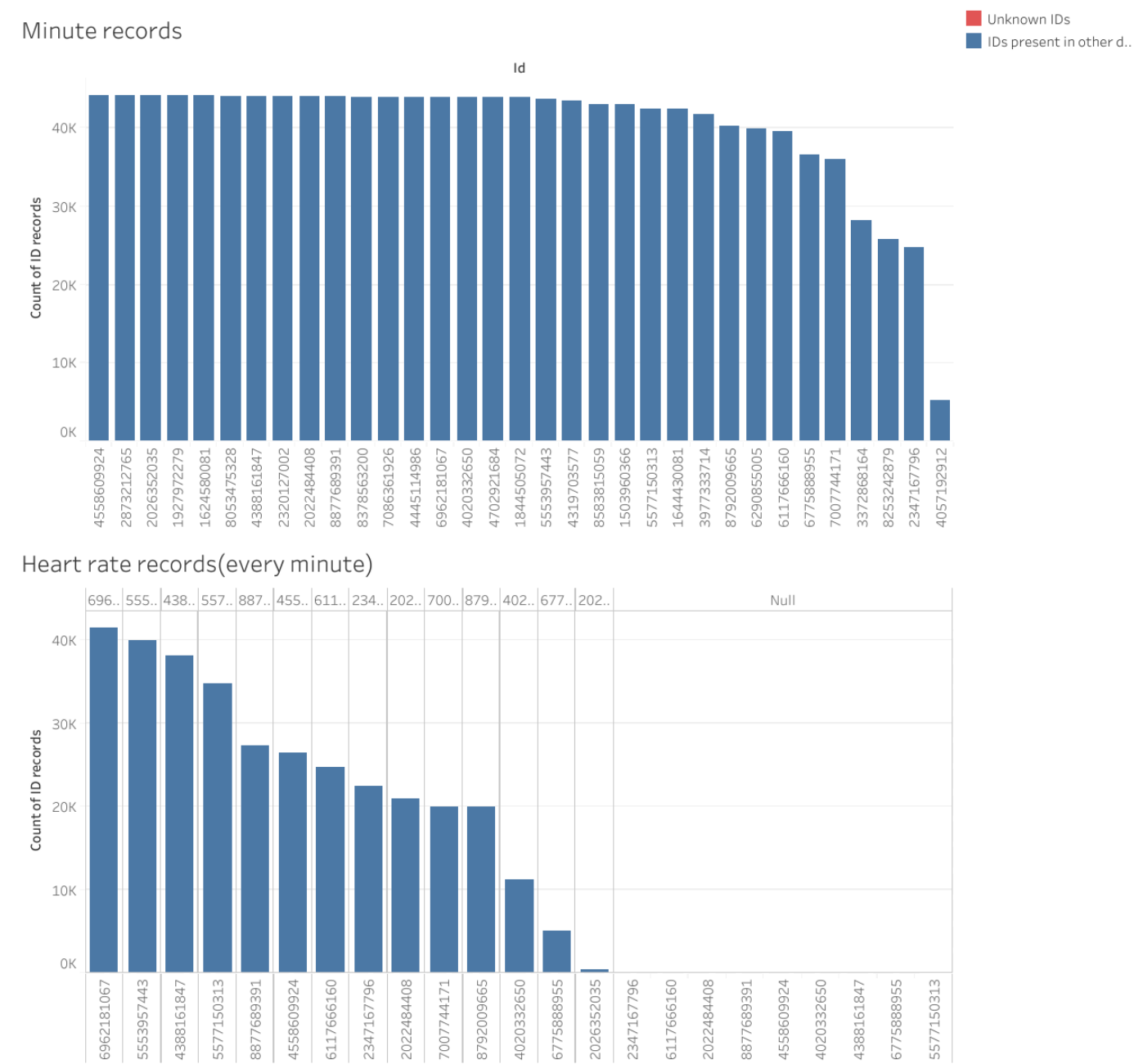
Data Quality Assessment

FitBit analysis
by Vladyslav Yakymchuk

The goal of this section is to demonstrate the overall reliability of the dataset and prevent unnecessary questions or misunderstandings.

The charts below show how many minute-level records each ID contains.

In the **Heart rate records** chart, the IDs shown as **Null(on the top)** are not present in the Minute Records dataset, but they each contain only around 50–60 entries, so they are not considered significant.

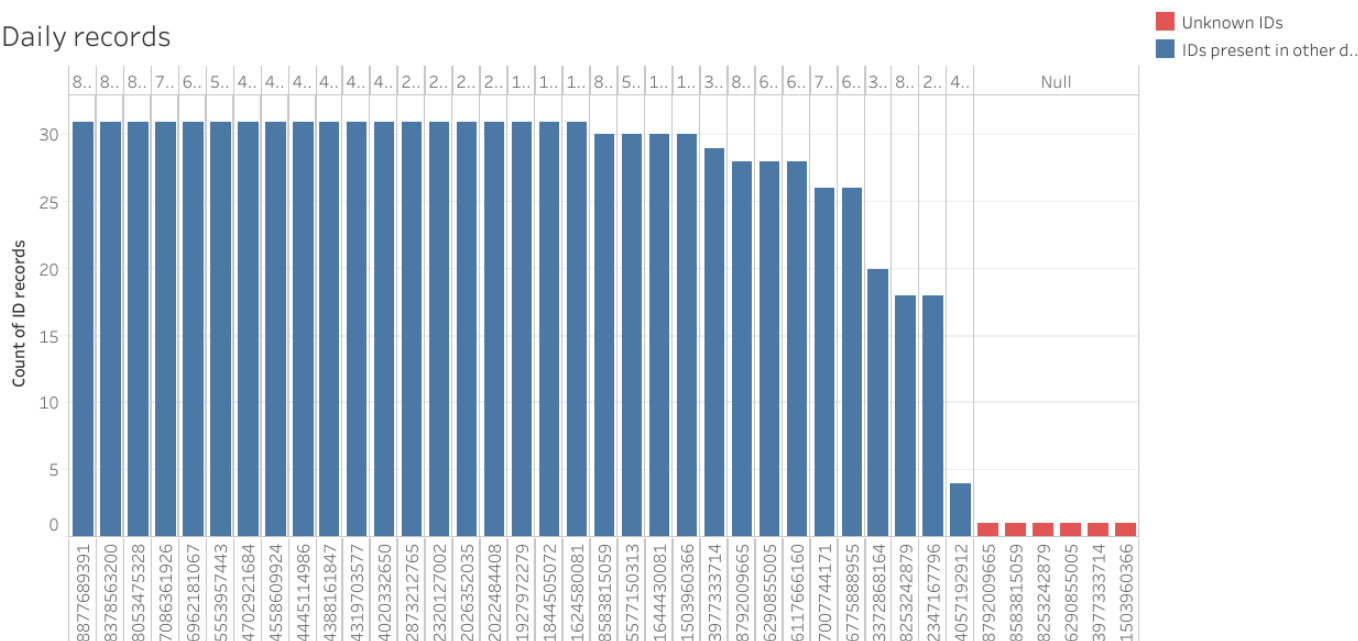


The next two datasets were already aggregated by day at the source, so each entry represents daily-level information rather than minute-level detail.

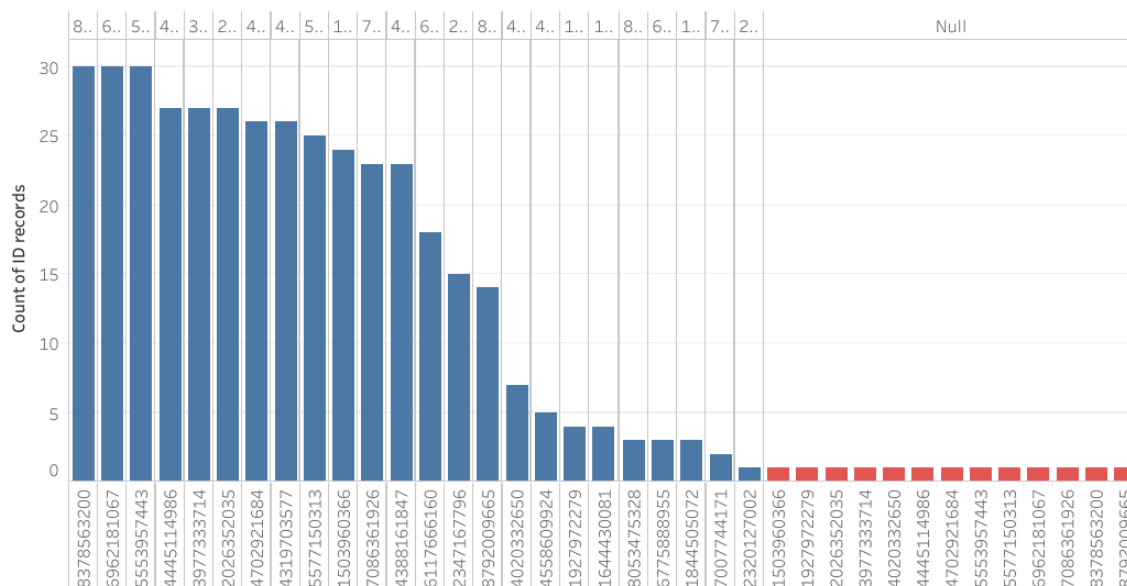
Here we again see IDs that are not present in the minute-level records, but these are insignificant because they typically contain only a single entry.

You can also observe that most IDs have all 30 daily records (one per day), while some have slightly fewer. In the Sleep Records dataset, the coverage per ID is somewhat lower.

Daily records



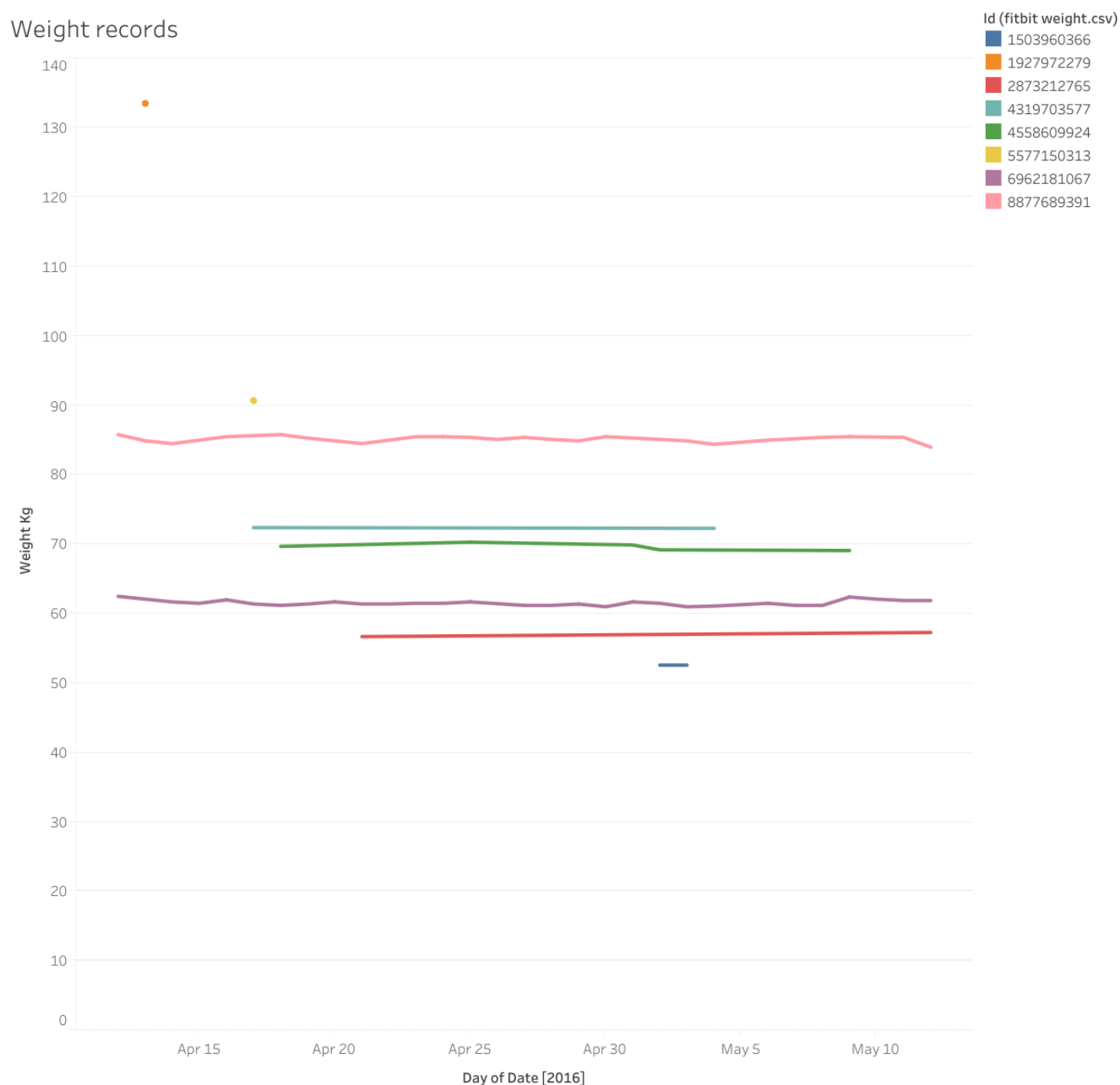
Sleep records(daily)



This has minimal impact on the subsequent findings, since the analysis relies on average values rather than totals. However, it may indicate that some individuals have slightly less accurate metrics due to having fewer data points.

The dataset also included daily weight measurements. However, I decided not to include this metric in the main report, as only two IDs have nearly complete daily coverage. A few other IDs contain only a small number of weight entries, which makes it impossible to assess whether their activity levels had any meaningful impact on weight.

The chart showing this is shown below.



I also aggregated some scatterplots not by 1 minute, but by 5 minutes. I mean, to find the correlation between, for example, calories and intensity (shown below), I use aggregation not by 1 minute as it is done in the csv file, but by 5 minutes. This is because calories can be burned not during intensity, but after. In this case, there is practically no difference, but it can still be useful in other cases.

I used five-minute aggregation only for the **calories vs. intensity chart**, as the others require simultaneous tracking of two variables.

A scatter plot showing the relationship between Average Intensity per minute (x-axis) and Average Calories per minute (y-axis). The x-axis ranges from 0.0 to 1.0, and the y-axis ranges from 0.0 to 4.0. A solid black regression line indicates a positive correlation. A vertical dashed line is drawn at x=0.0.

A scatter plot showing the relationship between Average Intensity per minute (x-axis) and Average Calories per minute (y-axis). The x-axis ranges from 0.0 to 1.0, and the y-axis ranges from 0.0 to 4.0. A solid black regression line indicates a positive correlation. A vertical dashed line is drawn at x=0.0.

	Minute of Activity Minute	Minute of interval5min	
April 12, 2016 at 12:00 AM	April 12, 2016 at 12:00 AM	Abc	
April 12, 2016 at 12:01 AM	April 12, 2016 at 12:00 AM	Abc	
April 12, 2016 at 12:02 AM	April 12, 2016 at 12:00 AM	Abc	
April 12, 2016 at 12:03 AM	April 12, 2016 at 12:00 AM	Abc	
April 12, 2016 at 12:04 AM	April 12, 2016 at 12:00 AM	Abc	
April 12, 2016 at 12:05 AM	April 12, 2016 at 12:05 AM	Abc	
April 12, 2016 at 12:06 AM	April 12, 2016 at 12:05 AM	Abc	
April 12, 2016 at 12:07 AM	April 12, 2016 at 12:05 AM	Abc	
April 12, 2016 at 12:08 AM	April 12, 2016 at 12:05 AM	Abc	
April 12, 2016 at 12:09 AM	April 12, 2016 at 12:05 AM	Abc	
April 12, 2016 at 12:10 AM	April 12, 2016 at 12:10 AM	Abc	

I also shifted my sleep day back by one day to match the activity date, because **sleep usually occurs the next day**. Here is a screenshot from the ipynb file:

Edited

While working with visualization in Tableau, I noticed a problem: the data on daily activity does not quite match the daily data on sleep. That is, daily activity occurs on a specific day, while sleep usually occurs on the following day (00:00 - 23:59). Therefore, I will make an adjustment so that it is possible to aggregate correctly by day.

```
1: df_sleep['activity_date'] = df_sleep['SleepDay'] - pd.Timedelta(days=1)
```

The data taken from: <https://www.kaggle.com/datasets/arashnic/fitbit>

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Charts generated via *Tableau Public*

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